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Case Study

A rare and severe manifestation of Jaccoud's Arthritis involving bilateral lower limbs

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Abstract:-

Rheumatic fever is a systemic disease involving joints, heart, pleura, skin and subcutaneous tissue and basal ganglion. A rare manifestation of rheumatic fever Involving small joints of hands resulting in prolonged disability and deformity called as Jaccoud's Arthritis (I). Recognition of this entity is important especially for differentiation from JRA (Juvenile Rheumatoid Arthritis) as long-term streptococcal prophylaxis will be required for the former

Introduction

No case reports have been published till date involving bilateral ,combined both hand and foot involvement .We present here a rare manifestation of rheumatic heart disease Involving small joints of hands and small joints of foot resulting in prolonged disability and deformity

Case Report

A 37-yr-old woman was admitted to our department in MAY 2014 with an 10-yr history of recurrent acute mono/oligoarthritis involving the wrists, MCP joints and ankles and lasting only 1 or a few weeks. She was a known case of rheumatic heart disease diagnosed 10 years back. For the last 5 yr the patient had complained of progressive deviation of the great toe

On admission to our department, history revealed worsening breathlessness, fever and swelling of the feet since one month. sore throat, palpitation, drug intake, epistaxis, rash and dysuria were denied. On examination patient had tachycardia. Patient had signs of acute decompensated heart failure. Cardiac auscultation prompted the diagnosis of severe mitral stenosis and severe mitral regurgitation with mid diastolic murmur and 5/6 pansystolic murmur at mitral area.

On investigation, throat swab for streptococci hemolyticus was negative. Blood culture was negative. Hemogram showed polymorphonuclear leucocytosis with ESR of 40. Mantoux test was negative. Rheumatoid factor and ANA were also negative. ASO titer repeated twice was 400 Todd units. CRP was positive.

The only findings of note in our extensive laboratory tests were a high erythrocyte sedimentation rate (ESR) (40 mm/h), ASO titer of 400 Todd units and a high C-reactive protein concentration (8 mg/dl). In particular, the following were normal or negative: rheumatoid factor, Waaler–Rose reaction, serum glucose, serum calcium, serum phosphate, serum uric acid, hemoglobin, iron and ferritin concentrations, antistreptolysin-O antibodies, immunoglobulins, cryoglobulins, antinuclear antibodies, anti-DNA antibodies, anti-ENA (extractable nuclear) antibodies.

X-ray chest showed cardiomegaly. ECG showed a heart rate of 124/minute with atrial fibrillation with fast ventricular rate.

2D echocardiogram concluded severe mitral stenosis with severe mitral regurgitation. Pulmonary systolic pressure was 70 mmHG. Reduced left ventricular function.

Examination and X-ray of the foot revealed complete subluxation of the proximal phalanx of the great toe and partial subluxation of the second and third toe. Severe limitation of movement in these joints was noted. X-ray of the hand revealed partial subluxation of the distal interphalangeal joint of the thumb bilaterally. Bilateral Metacarpophalangeal joint subluxation of the little finger was also noted. There was no destruction of bony joint surfaces.

Discussion

Joints are involved in 70% of initial attacks of rheumatic fever¹. The arthritis is classically migratory (flitting) lasting a day or so in one joint and then disappearing and involving the other joints. It classically involves major joints especially knee joint, elbow joint and ankle joint. Temporomandibular and sternoclavicular joint involvement is not described in rheumatic fever.

Very rarely it has been described in rheumatic fever with minor or some deformities of hand that can occur which may be persistent and are known as Jaccoud's Arthritis.^{2,3}

References

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Figures



Fig 1. Images showing severe bilateral lower limb with Jaccouds arthritis

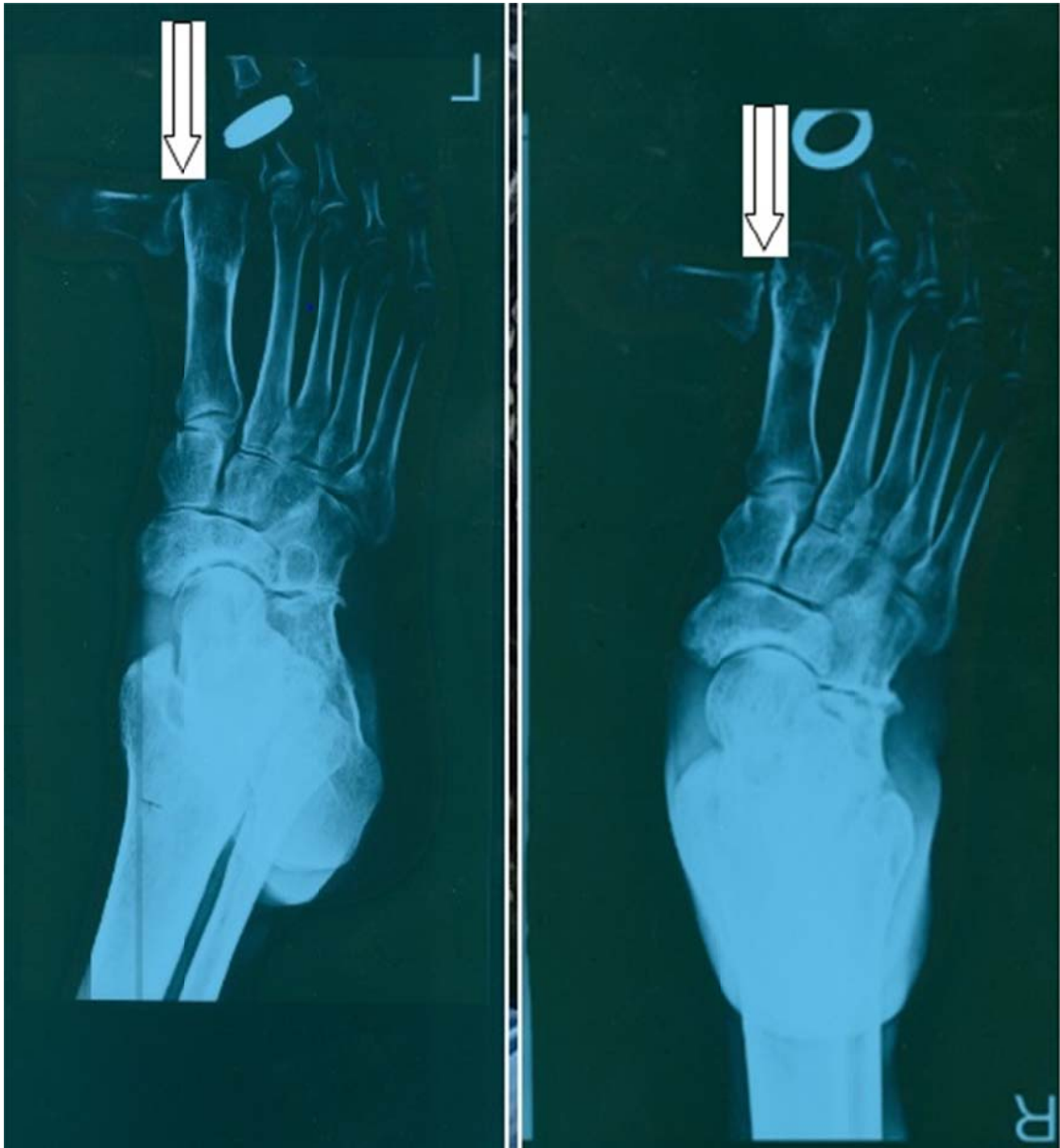


Fig 2: X-Ray of the foot showing subluxation of Phalanges (Marked by arrow)

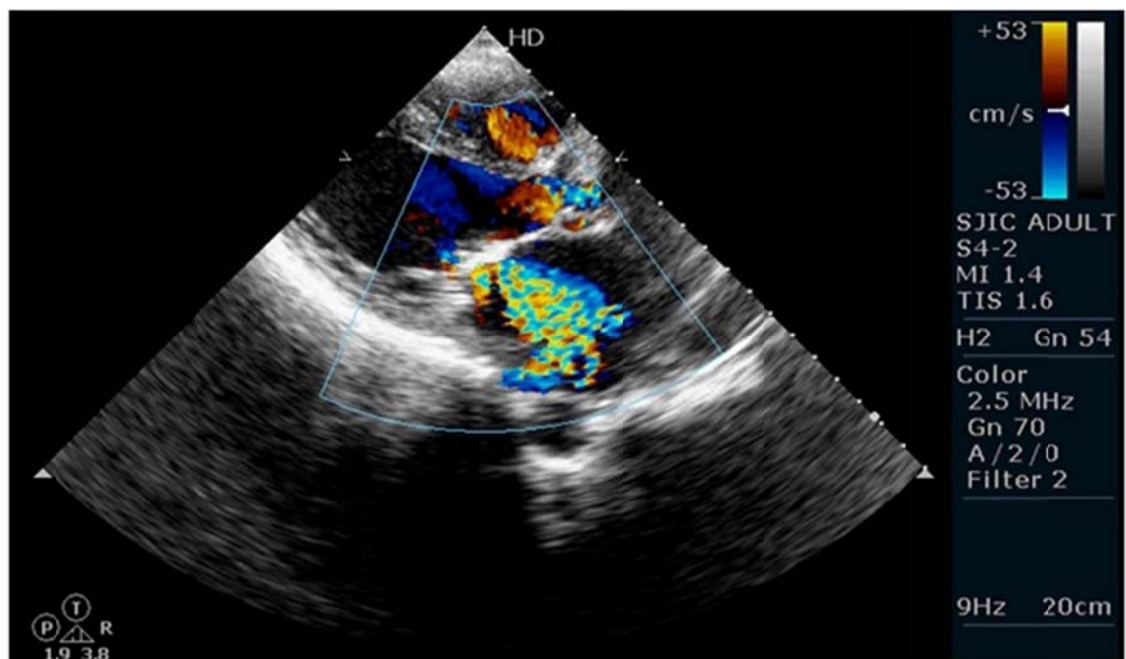


Fig 3: 2D Echo Plax view showing thickened leaflets, severe mitral stenosis and color mode showing mitral regurgitation